Project Proposal

Using Neuroevolution for Intrusion Detection

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Problem Statement:

Current intrusion detection systems are limited in their efficiency against a constantly changing cyberspace of malware, bad actors and zero-day attacks [1]. Many anomaly-based intrusion detection systems (AIDS) use traditional machine learning techniques such as clustering and decision trees to identify malicious network activity, but there has not been extensive exploration of neural networks evolved using genetic algorithms (neuroevolution) as a tool for intrusion detection [1]. Neuroevolution is a promising addition to the cybersecurity field for its ability to process large datasets, consider real-time data and retrain and adapt to different environments and contexts, and in other fields has repeatedly outperformed solutions derived via traditional machine learning techniques [3].

Stakeholders:

Primary Stakeholders: Dr. Akbas and his research team.

Secondary Stakeholders: Society.

Proposed Solution:

Using a neural network evolved using a genetic algorithm to create an intrusion detection system with the ability to recognize anomalies in the network.

Semester 1:

The overall goal of this project is to evolve neural networks using HyperNEAT (a genetic algorithm) for the purpose of identifying intrusions in network traffic. We will evolve and test our neural networks using one or more open-source intrusion detection datasets. These datasets are composed of network related metadata and packet information,

including data representative of both normal network behaviors and malicious activity, allowing us to evolve a neural network with the ability to identify genuine malicious activity within normal network traffic. The resulting neural networks will output a classification of the input data as either normal or suspicious [3].

In evaluating the fitness of our neural network, we will use three different fitness algorithms. The first will calculate fitness based on optimization, assessing the output of the neural networks on accuracy and precision. The second will calculate fitness based on the novelty of the behavior demonstrated by the neural network, which has been shown to achieve higher performance on dome tasks [4]. The third will seek a balance between optimization and novelty. In this way, we can explore how much of an effect novelty has on the ability of our neural network to perform intrusion detection tasks successfully.

We will present our results, including the evolutionary history and fitness performance over generations, in a visual summary through graphical and tabular means.

In the project we will constrain our source of training and testing data only to open-source data sets so others will more easily be able to replicate our results. Due to both limited resources and time constraints, we will not be attempting to use a real-time network as a source of data.

Semester 2:

There are a few different paths that we may pursue in semester 2, depending on our rate of progress throughout semester 1.

One possible avenue for expansion is the use of different genetic algorithms for evolving neural networks for intrusion detection, and comparison with our HyperNEAT results from semester 1. In doing so, we can better assess neuroevolution's overall potential for creating AIDS that are both optimized and explore novel solutions.

Another possible way to expand the scope is to create a graphical user interface (GUI) that allows a user to more easily utilize the neural network to detect anomalies in their own data sets. Our project will not produce a system that triggers an active protection against malicious activity but would simply categorize network behaviors as normal or

anomalous. The primary purpose of creating a GUI would be to increase the ease of access to this software, and ideally make it a more marketable product.

Proposed Project Budget:

The team does not foresee needing monetary assistance for semester 1; depending on the project's future growth, some Cloud storage may be needed in semester 2.

References:

[1] K. Ansam *et al*, "Survey of intrusion detection systems: techniques, datasets and challenges," Cybersecurity, vol. 2, (1), 2019. Available: https://www.proquest.com/scholarly-journals/survey-intrusion-detection-systems-techniques/docview/2259210177/se-2. DOI: https://doi.org/10.1186/s42400-019-0038-7.

[2] F. P. Such, V. Madhavan, E. Conti, J. Lehman, K. O. Stanley, and J. Clune, “Deep neuroevolution: Genetic algorithms are a competitive alternative for training deep neural networks for reinforcement learning,” 2018, arXiv:1712.06567v3.

[3] B. Su, R. Li and H. Zhang, "Evolving Deep Convolutional Neural Network for Intrusion Detection Based on NEAT," 2020 23rd Int. Symp. Wireless Personal Multimedia Communications (WPMC), Okayama, Japan, 2020, pp. 1-6, DOI: 10.1109/WPMC50192.2020.9309451.

[4] J. Lehman and K. O. Stanley, “Novelty Search and the Problem with Objectives,” Genetic and Evolutionary Computation, pp. 37–56, 2011, DOI: https://doi.org/10.1007/978-1-4614-1770-5_3.